



AuraLink

Your home, aware of you.

Two products for the Apple home: AuraLink for Apple Watch, and Room Sense

Two gaps in the Apple home



CONTROL

Names fail

- Six lamps, all called "Lamp"
- Guests don't know any device names
- The phone is in another room



CARE

Worry from afar

- Parents living alone, at risk of falls
- Cameras feel invasive in bedrooms and bathrooms
- A watch isn't worn in the shower or overnight

AuraLink for Apple Watch



Raise your wrist

AuraLink surfaces at home.



Devices rise up

The ones you're facing, closest first. No aiming needed.



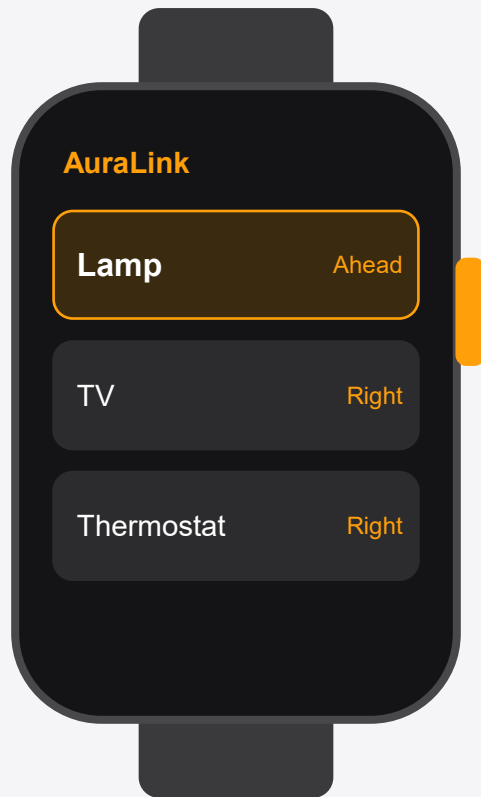
Double-tap

Opens the top device. Or tap any in the list.



Turn the Crown

Brightness, temperature, blinds, volume. A haptic tick every step.



Feel every step: three tiers of haptics

Tier	What you feel	Where	Extra cost
Software	A tick per step: 5% light, 1°F thermostat, 10% blinds	All recent Apple Watches	None
Braked Digital Crown	Real resistance near the limits, and a hard stop at them	Premium models, such as Ultra	about \$5 per unit
Directional haptic band	4 motors tell you which way a device is: ahead, left, right, behind	Optional band	\$129 price, about \$30 to make

Why 4 zones, not 8: people reliably tell apart only about 4 vibration spots around the wrist. Design to the physics of touch.

Room Sense: care without cameras



Falls

A fast drop to the floor, then staying down, alerts family. Getting up cancels it.



Presence

Knows someone is there even when sitting still, from breathing micro-motion.



Sleep breathing

Breaths per minute while lying still, as a wellness trend in Health.



No camera. No images. A small plug-in 60 GHz radar measures only distance and motion, and processes it on the sensor. It covers the times a Watch isn't worn: the shower, and overnight.

Telling a fall from sitting down



A FALL ALERT NEEDS ALL THREE

1 Fast drop

Body moving down faster than 1.4 m/s

2 To the floor

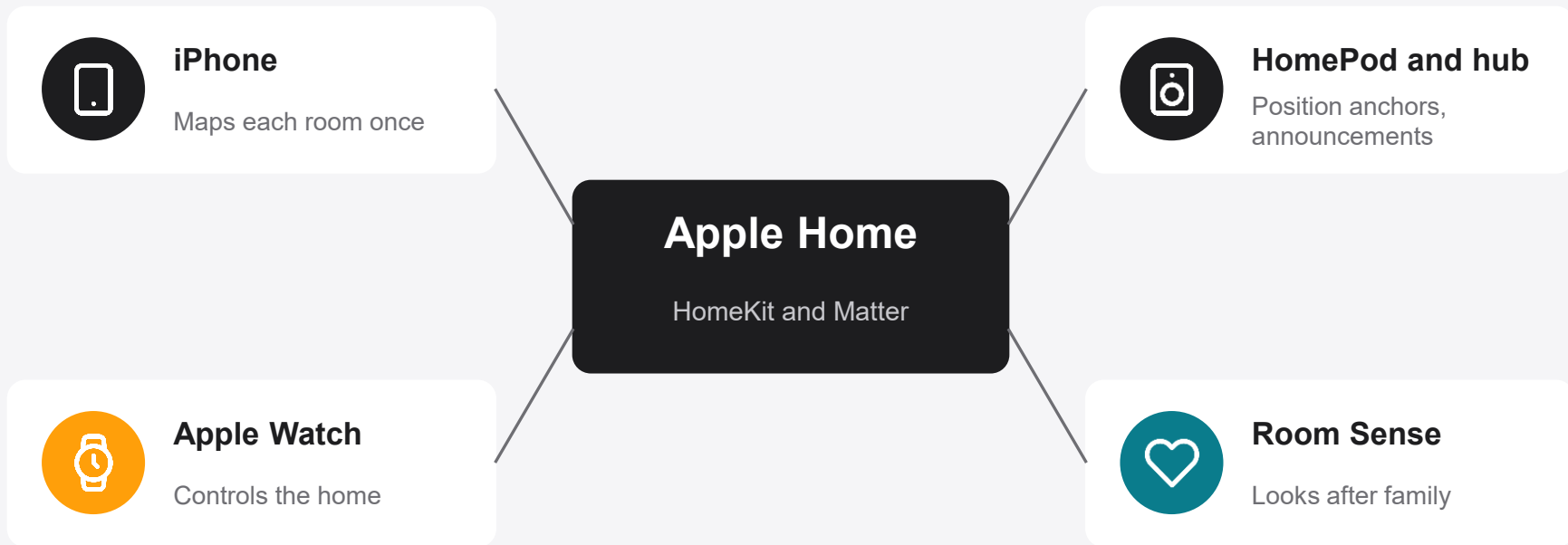
Body centre below 0.45 m

3 Stays down

10 seconds without getting up

Sitting down fast stops at seat height. Bending down comes back up. Lying in bed is slow. None of them alert.

One foundation, every device more valuable



Together: up at 3 a.m.? Room Sense knows, so your Watch puts the dim hallway light at the top.

Engineering targets

AURALINK FOR APPLE WATCH

90%

Right device on top

<0.3 s

List updates as you turn

$\leq 5\%$

Battery impact

90%

Band directions identified

ROOM SENSE

95%

Real falls detected

<1

False alert per home per month

± 1

Breath per minute during sleep

0

Images captured

Targets, to be validated in pilots and studies.

The business case

AuraLink for Apple Watch

Extra Watch upgrades ~\$400M

Development \$20M~\$30M

Base case ~\$370M a year

Room Sense

\$149 sensor, ~70% margin ~2.5M units

Revenue ~\$373M

Development \$30~\$50M

Base case ~\$323M a year

About \$0.8B a year combined, before HomePod and hub pull-through. Room Sense breaks even at ~385,000 units.

All figures are illustrative estimates, not Apple data. Watch figures assume 30M units a year and a 3% upgrade lift.

What's built today



github.com/vivekselva91/Auralink

- Swift core with unit tests: facing list, Crown profiles, 4-zone haptics
- iPhone room setup that sends the map to the Watch
- watchOS app: facing list, double-tap, Crown control
- Room Sense algorithms, simulator, and tests in Python



Concept animation: [\[add link\]](#)

ROOM SENSE, SIMULATED (50 RUNS EACH)

Scenario	Alerts
Fall, stays down	50 / 50
Fall, gets up within 5 s	0 / 50
Sits down fast	0 / 50
Bends to pick up	0 / 50
Lies down in bed	0 / 50

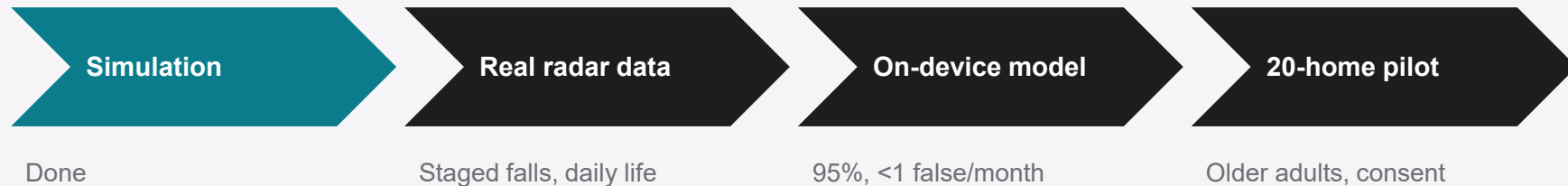
Simulated data only: it shows the logic works as designed. Real accuracy will be lower and must be measured on recorded radar data.

Roadmap: prove each with people

WATCH



ROOM SENSE



Risks I'm managing

Indoor compass drift

Rank a list, don't aim; $\pm 60^\circ$ view

Watch battery

Brake and band in premium tier only; $\leq 5\%$

False fall alarms

10 s confirmation; cancel on getting up

Medical claims

Safety and wellness only; regulatory review

Why I built this

It's how I approach sensing products: start from the physics, design for power and haptics, make the business case, and test with people early. I'd value 15 minutes of your feedback.